HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed **distributed backend services in Java 11/17 and Spring Boot** (REST APIs, microservices) for a banking payments platform, using **design patterns** (circuit breaker, retry) for **reliability**.
- **Scaled backend services horizontally and vertically** on **AWS (EC2, Lambda)** with Docker, stateless design and load balancing, cutting p95 API latency by **25%** during month-end peaks.
- Improved **system performance and memory management** of production JVMs via heap/thread-dump analysis, GC tuning and connection-pool sizing in **Dynatrace**, ending recurring OutOfMemory incidents.
- Applied **parallel computing principles** with Java concurrency (ExecutorService, CompletableFuture, thread pools) to batch reconciliation and ETL jobs, cutting processing time by **40%**.
- Led **data migration and replication** across Oracle, PostgreSQL and MySQL, **sharding data by key range** into parallel, restartable batches with checksum validation: **1.5 TB moved with zero data loss**.
- Implemented **authentication/authorization protocols** (Spring Security, OAuth2, JWT) for user and service-to-service traffic, with role-based access control aligned to GDPR.
- Served on the **on-call rotation** for P1/P2 incidents, driving **root-cause analysis** in Splunk and CloudWatch and shipping permanent fixes, holding **99.9% SLA** and cutting repeat incidents by **35%**.
- Practised the **full software development life cycle**: coding standards, **code reviews**, Git source control, Jenkins build processes, JUnit testing and operations; presented in **design reviews**.

SKILLS

Languages – Java 11/17 (concurrency, JVM performance tuning), Python, SQL, Bash.

Backend – Spring Boot, Spring Cloud, Spring Security, Hibernate/JPA, REST APIs, microservices.

Data Stores – PostgreSQL, MySQL, Oracle, Redis; replication, sharding, data migration, query tuning.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins CI/CD, Git, Linux.

Operations – On-call (P1/P2), root-cause analysis, Splunk, Dynatrace, CloudWatch, SLAs, runbooks.

Concepts – Distributed systems, horizontal & vertical scaling, high availability, low latency, memory management, parallel computing.

PROJECTS

Distributed In-Memory Caching Layer for High-Throughput Services

- Built a **Redis OSS**-backed distributed cache (cache-aside, TTL eviction) in front of Spring Boot APIs, serving hot reads in **under 5 ms** and cutting database load by **60%**.
- Implemented **data sharding** with consistent hashing and primary-replica **replication** with automatic failover for **high availability**.
- Added online key migration so cache nodes **scale out horizontally** without downtime; benchmarked **vertical scaling** by instance size.
- Secured access with TLS and token-based **authentication/authorization**; published latency, hit-rate and memory metrics to CloudWatch alarms.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)